LESSON PLAN

PYRO TECHNIQUES (34)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Control system, type and requirements
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Control system, type and requirements
5. Specific Objectives	34.1.1.1 Trainee will be able to explain various terms on guided weapon.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of guided weapon terminology diagrams; Safety: No live equipment handling; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a control system in a guided weapon?
(b) List three common types of guidance systems used in missiles.
(c) Define the term 'command guidance' in your own words.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Control System • Basic Components Overview • Signal Transmission Path • Actuator Functionality • Feedback Loop Mechanism	L & Ds	PPT, IB	15 min
(b)	• Types of Control Systems • Open Loop vs Closed Loop • Proportional and Integral Control • Derivative Action Effects • System Stability Criteria	L & Ds	PPT, PDF	20 min
(c)	• Specific Requirements Analysis • Response Time Standards • Accuracy and Precision Needs • Environmental Hardening • Redundancy Protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a closed-loop control system corrects errors during flight.
(b) Describe the specific requirements for a control system operating in high-altitude environments.
(c) Differentiate between open-loop and closed-loop systems based on feedback presence.

12. Home Assignment and Follow Up	Prepare a brief note on the power supply requirements for the control system discussed in LP 08, highlighting the interface with the control unit.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Power supply system –purpose and requirements	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min